

AI Enhanced Password Security Tool

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1 ABSTRACT

With the advent of computers in the 20th century, humans could now offload much of our computational overhead to machines. AI is now doing the same for our conscious and cognitive abilities. This opens up novel avenues with regards to Cybersecurity. Perhaps the most significant affect will be felt in the realm of intelligent target profiling attacks, which have so far required individualised effort from the attacker to execute. However, AI can now perform much of the same profiling a human could. This poses a significant threat, particularly for specialized brute force password attacks. This project built a tool which would explore a mode of these attacks, and also implemented potential mitigation strategies for it. I found that as of now, the current bottle neck of resources and lack of credential related training data makes it unlikely for AI to pose a major threat to specialized brute force attacks, but it is only a matter of time before it renders most user generated passwords useless.

2 INTRODUCTION

In the past 10 years, we have seen a dramatic rise in the measures taken by various platforms to increase password security, such as increased and more complex password requirements, and often including double factor authentication. This is a symptom of the increased risk that user credentials have come under due to increasingly sophisticated password attacks by hackers. However, the primary danger towards user generated passwords does not come from hacking, but

from various social engineering techniques. A common technique that falls under this category is phishing. This is a technique where a user is tricked into revealing personal information about themselves by thinking that they are entering their credentials on a legitimate page, when in reality it is a copy made by a hacker. Another social engineering technique is to try to narrow a user's password down to a select few options by gathering information about them and attempting to crack their password through that. For example, using the forget password option and then using the extracted information regarding the user to answer their security questions, and accessing their account. Yet another security risk is that of raw brute force attacks. This would be the most computationally intensive option, and is currently unfeasible. However, as we'll discuss, it can become more and more dangerous with the recent developments in AI technology.

In the past, these techniques have required extensive effort on part of the attackers. Although phishing could be accomplished by simply sending out massive amounts of emails, it's often just a hit or miss. For example, an email sent out attempting to get people to sign in to their bank accounts, would not be able to tell which exact bank each user has. Thus the users would catch on to the trick. Also, regarding extracting specific information about users would require an individual to dive into a users data, either physical or virtual, and extract what information they could gather in order to infiltrate their accounts. As for brute force attacks, they are currently an ineffective way to hack into a user's account. This is due to most sign in pages having a limit to how many attempts a user can make before their account gets locked. However, this is still dangerous for the integrity of a user's security. Whenever we hear of large scale hacks and data leaks from major companies, the credentials are often encrypted and impractical to decode for the attackers, but they do have unlimited attempts to do so. The hackers would know if they got it right and would then be able to access the user accounts which didn't bother changing their passwords, or even access their other accounts with the same passwords. The problem for attackers so far has been that running brute force attacks is extremely impractical given the amount of possible combinations for most passwords. However, given what we know about the common and extremely predictable habits that users have when it comes to their passwords,

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we can narrow down the candidate passwords to a smaller subset of maybe a few thousand likely combinations.

With the advent of AI, this job is made all the more simpler. An attacker does not need to sift through a user's info to find info to form social engineering, or specialised brute force attacks on them. Attackers can now offload the cognitive and conscious pattern recognition and profiling to an AI tool that looks for certain information and patterns within a user's profile to now specialise these attacks.

My project's aim was to build a tool that would serve as a proof of concept, which explores the ways in which AI can be used to facilitate these attacks. Specifically, in how it can automate the cognitive overhead and scale them to be delivered in a specialised manner en masse to a user base. This inquiry would be made through implementing these attacks in a rudimentary social media site, and would be explored through use the perspective of mitigation. The main strategy my project employs to mitigate them is to simulate these attacks and warn the user if they are under any risk.

3 RESEARCH PROBLEMS

The goal of this project is to build a tool that demonstrates a potential way in which AI can be used to replace the conscious human element in Cybercrime. Particularly, Cybercrime related to password security. The assumption is that AI will largely lead to automating much of the intelligent target profiling (ITP) that was required in cyberattacks of a targeted and specialized nature. With automation, would also come scaling up these attacks. This would of course require automated mitigation strategies as well, and this project aims to explore the "think like the enemy" approach to build these mitigation tools. Currently even the best computers cannot perform raw brute force attacks on modern day passwords to any significant affect. However, a hostile government agency could have the resources required to utilize AI to perform specialized ITP attacks.

The questions I wish to answer through my project are:

- (1) What are the ways in which AI could be used to conduct attacks on passwords?
- (2) What are the possible bottlenecks and limitations of AI powered password attacks?

4 RELATED WORKS

Most of the research related to this project related to two major topics. The first being the underlying patterns and habits that people have when forming their passwords. The second, being research that relates to exploring the way AI could impact common cybersecurity problems. I shall now discuss some of the studies in more depth.

"Artificial Intelligence-Based Password Brute Force Attacks": This paper builds a tool which utilises AI to crack a user's password. However, it requires a directory of a user's prior passwords to do so. This tool demonstrates that AI can significantly enhance brute force attacks on passwords. However, it requires the use of a user's prior passwords in order to crack the new one. Whereas my project aims to use information about a user from their social media posts to specialise the attack. [1]

"Password strength: An empirical analysis": This study analyses and evaluates a number of brute force attacks on user passwords. However, all of its methods are still in the realm of the computational overhead which would make it infeasible to be scaled up to be used in a large scale attack. [2]

"Password Security: An Empirical Study" and "The importance of social identity in password formulations": Both these studies deal with a similar theme of identifying underlying habits and patterns that users demonstrate with regards to their password habits. These studies are critical to my project due to them showing in what way my project tool must mutate and transform the data that it gathers in order to narrow down the list of possible passwords to the most likely contenders. The studies do however lack the consideration of the extra requirements that most passwords have nowadays. Particularly with regards to special characters, numbers and capitalization. [3][4]

"The Emerging Threat of AI-driven Cyber Attacks: A Review": This study discusses how attackers can use intelligent target profiling to carry out specialised attacks against users. Specifically, the ITP is carried out with the help of AI. This study's meta analysis in its research phase is quite relevant. It attempts to synthesise and categorise the current literature regarding the topic of AI-driven cyber attacks with regards to pattern recognition, however it does not demonstrate any one specific method of forming profiles on users. [5]

"Cybersecurity Threats Based on Machine Learning-Based defensive Technique for Password Authentication": This paper deals with keyboard stroke data breaching. They attempted to mitigate the breaching by scrambling the keystrokes input data, however the machine learning algorithm they built was able to successfully decode the data more than 90 percent of the time. This demonstrates how AI has the potential to resurrect cybersecurity threats that had previously been solved through static solutions. [6]

5 IMPLEMENTATION METHOD AND RESULTS

5.1 Development environment

The development environment for this project consisted of the following:

- **OS:** Windows 10
- **Programming Language:** Python 3.12.2, chosen for its ease of string manipulation
- **Text Editor:** VScode
- **Testing Environment:** VScode Terminal
- **Libraries:** Levenshtein, Itertools, and openai
- **Other Requirements:** GPT API access, which is a paid service

Important note: Install the libraries using 'pip' before running the code on your system.

5.2 Code organisation

The code resides entirely within the `AIpassword.py` file, while data concerning accounts and posts is stored and retrieved from `accounts.json` and `posts.json`, respectively.

5.3 How to run

- (1) Install Python 3.12.2.
- (2) Install the dependent libraries such as `itertools`, `openai`, and `levenshtein` using "pip."
- (3) Ensure that `Accounts.json` and `posts.json` are in the same directory as `AIpassword.py`.
- (4) Run the code using "python3 `AIpassword.py`", ensuring you are in the same directory as `AIpassword.py`. (Note: Depending on your system's environment variables, you might need to run "python `AIpassword.py`" instead.)

5.4 Dataset

In this context of this project, the dataset can be defined as user accounts, and posts related to those accounts. There are clearly some ethical and technical issues with obtaining any realistic data regarding this, so I simply tested the tool by synthesising the profile of an average social media user, and created posts that would pertain to day to day life. Such as talking about daily activities, relationships and interests. The data is stored in files using JSON format, and is operated on through lists in the program itself. Some of the pre-processing involved with the data is keyword extraction, join the keywords and mutating them to meet certain criteria. This is discussed in further detail in the "AI powered algorithm" section of the paper.

5.5 Preliminary work

The first step of developing this tool was to conduct some background research regarding the underlying patterns and habits that users have related to their passwords. This was necessary because the tool will be gathering keywords from a user's social media posts, and then has to form a list of likely candidates for the user's passwords. The research would help determine what keywords to look for in the users posts, for example most family members names and hobbies. The second reason was to learn how to mutate those terms to meet the necessary requirements for the password through a minimal amount of candidate passwords. For example where and which special characters to append to the keywords.

The second step was to build the simulated social media environment in which my tool would run. I decided to keep it simple and build a bare bones social media platform clone which would operate in the terminal. It would have the options to:

- Sign in
 - Make posts
 - View posts
 - Check password strength
 - Change password
- Create account
 - Create Username
 - Create Password which meets a certain requirement

5.6 The Algorithm

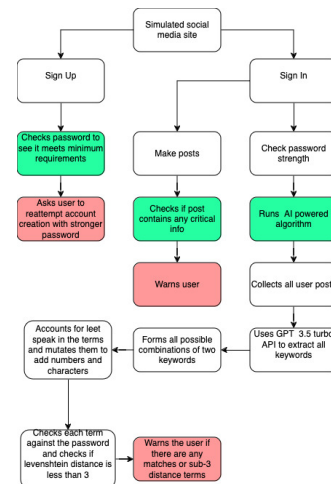


Figure 1: Enter Caption

The password security enhancing tool would operate at three different points in the simulated social media site:

- (1) **The point of creation:** When the user signs up for an account, their username is checked for uniqueness. Then, their password is checked if it meets all the requirements for a strong password. These requirements are the basic requirements we find in most sites nowadays. Such as a minimum character length requirement of 8, at least 1 special character, and a number.
- (2) **Making posts:** When the user makes a post, this post is checked to see if it contains any substring which could be a substring of the user's password. This feature can be further expanded to check if the users posts contain any information that could make them prone to social engineering attacks. Such as naming other social media websites they use, or giving the name of their bank or courier service.
- (3) **Manually check password strength:** In the menu for the signed in user mentioned earlier, there is an option for the user to manually check the strength of their password. This is done through an AI powered procedure described in the next section.

5.7 AI powered algorithm

The AI powered algorithm works as follows:

- (1) First, the users' posts are collected into a single list of strings and this list is passed to the GPT-3.5-turbo-0125 model API. The API is prompted to extract any information regarding relationships (including family and friends), names, places, pets, and hobbies. The API call also initiates a return of the results in a JSON format, so that the results can be easily extracted and operated on in the Python code.
- (2) Now, with the relevant terms all gathered into a single list, the algorithm runs a series of procedures on the keywords to mutate them into possible and likely password candidates:
 - (a) A list of terms of all possible two combinations of any two words is formed.
 - (b) Each term in the list is then checked for any "leet speak", which is when letters are replaced with numbers or characters, such as replacing 'E' with '3'.
 - (c) Finally, the terms are then appended and prepended with numbers and characters that are most commonly used by users when trying to form a password that meets the requirements.
- (3) Check each term in the list against the user's password. If the password is a match, then the algorithm successfully cracked the user's password and they are warned. Otherwise, the Levenshtein distance is calculated between the term and the user's password. If the distance is less than 3, then the term is added to a list of terms which came close to guessing the password, and the

user is shown the list with a warning of a security risk to their password.

5.8 Results

Upon testing the tool, all three steps of the security tool worked as intended. The basic point of creation password strength checker ensured that users' passwords met a certain requirement. The info detection tool at the point of making a post warned the user if their posts contained any substring from their password. Finally, the AI-powered algorithm successfully extracted the expected keywords from the list of posts for a user and mutated them to form a comprehensive list of password candidates. For the sample account I made, the password length was 16. The number of extracted keywords was 18 and after running the mutation functions, 18360 potential candidates were found for the password, of which 1 was an exact match. In contrast, if we were to run a brute force algorithm to test all possible passwords of length 16, there would be 3.72×10^{37} possible candidates.



Figure 2: Insufficient password when creating account

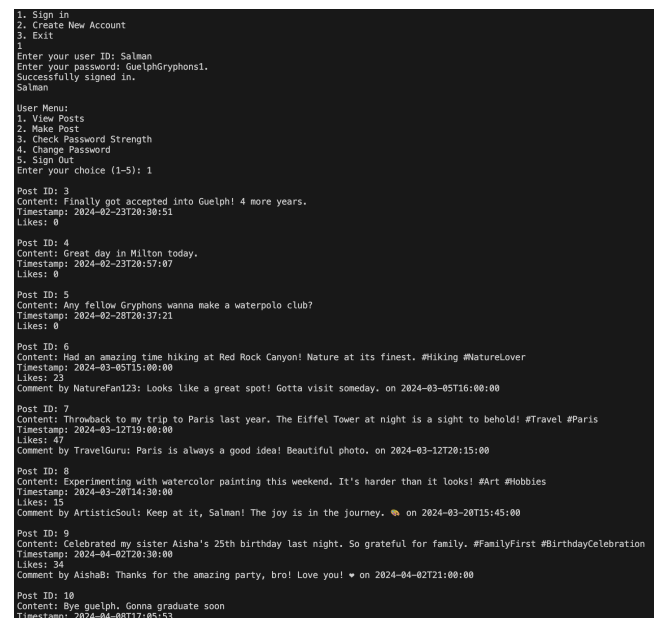


Figure 3: List of posts for a synthetic user

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User Menu:
1. View Posts
2. Make Post
3. Check Password Strength
4. Change Password
5. Sign Out
Enter your choice (1-5): 2
Enter your post content: Wonder if Guelph has any good bowling alleys
Warning: Your post contains a word that is also in your password. Are you sure you want to proceed? (yes/no): no
Post creation cancelled.

```

Figure 4: User is warned of potential security risk with post

```

User Menu:
1. View Posts
2. Make Post
3. Check Password Strength
4. Change Password
5. Sign Out
Enter your choice (1-5): 3
Keywords: 'Guelph', 'Milton', 'Gryphons', 'waterpolo club', 'Red Rock Canyon',
'Hiking', 'Naturelover', 'Paris', 'Eiffel Tower', 'Travel', 'watercolor pain
ting', 'Art', 'Hobbies', 'Aisha', 'FamilyFirst', 'BirthdayCelebration', 'snow
day'
Total possible combinations found: 18360
Exact match found: GuelphGryphons1.
This password is not strong enough.

```

Figure 5: User manually checks password strength and fails security check

5.9 RQ 1: AI Usage in Password Attacks

What are the ways in which AI could be used to conduct attacks on passwords?

In the process of building this tool, I learned that AI will most certainly open new avenues of cybersecurity issues. Particularly, with regards to password attacks and the issue of using intelligent target profiling (ITP) to design social engineering attacks to target users. I was using the GPT 3.5 Turbo model for this project, but some testing of the costlier GPT4 model demonstrated significant enhancements in keyword retrieval. Specifically, it would also return keywords that could be associated with the user. For example, if the user mentioned basketball and Toronto, it would infer that they are fans of the Raptors and mention some terms related to that. However, the real danger is if someone trains a machine learning model specifically on a real dataset of user accounts, posts, and passwords. This AI model would be specifically designed to learn and exploit the underlying patterns and associations between a users' posts and their passwords.

5.10 RQ 2: Bottlenecks and Limitations of AI in Password Attacks

What are the possible bottlenecks and limitations of AI-powered password attacks?

The potential for developing this type of offensive AI model is there and conceptually devastating. However, the main bottleneck lies in the resources required for such a tool. First, there is the computational overhead required to train these models. This would be a massive undertaking in terms of financial resources since it would not be able

to use any preexisting skeleton models available through OpenAI, Facebook, or Google due to their protective policies and safeguards. Secondly, and perhaps the bigger hurdle, is the lack of any substantial and accessible dataset which the model could be trained on. Most of the database leaks from social media sites, which are themselves rare, do not contain the passwords in plaintext, but rather their hashed versions. This is useless for the purpose of training a database for connecting posts to their passwords in plaintext.

6 CONCLUSION

At the concluding point of this project, I accomplished much of my initial goals. My tool successfully checks the password at all 3 points in the process of using the social media simulator. At creation, when making posts, and when the user manually wishes to do so. I also gained important insights regarding password security. This demonstrated not only the potential use cases of AI in the realm of password related cybercrimes, but also a potential mitigating strategy against such attacks. What I have concluded is that it is only a matter of time before user generated passwords become obsolete, especially when it comes to social media websites. The current bottlenecks of computation power and sufficient training data are simply a matter of resources. The benefit from being able to launch large scale devastating attacks makes it probable that some nefarious agents, perhaps a government agency, would be willing to make the necessary investments required to develop this tool. Ultimately, I think that user generated passwords will be rendered obsolete in the near future. The future of passwords and credential based applications will be some form of randomly-generated password keys. There are already numerous applications that do this, but they still operate as an accessory rather than an integrated and required part of the sign in process. Another possible solution could be bio metric logins becoming more mainstream.

7 FUTURE WORK

The goal of this project was to serve as a vehicle of inquiry regarding the novel avenues that AI opens up in the realm of password related cybercrime, and to get a grasp of its limits. As such, there are still many directions in which this tool could be fleshed out by further research and development. In my implementation, I used the GPT 3.5 Turbo model. However, using the more powerful GPT 4 model would have greatly enhanced the abilities of this tool, but it's bigger price tag is a bit of a hurdle given the scope of this project. Additionally, the tool could benefit greatly from having more comprehensive social engineering mitigation protocols. One pertinent one is modifying the tool to check the post a user is about to publish and ensure that it does not contain any

information that could be related to their security questions. The tool could also ensure that the user does not reveal information about other services that they use and likely have accounts for, such as banks, other social medias, or cellphone providers.

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